

Permutation-Cell Auditing

Target-free falsification of all-order retrain equivalence
in sequential machine unlearning

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Abstract

Sequential machine-unlearning services may process the same fixed deletion set in different orders while claiming that every terminal model approximates one common retraining result. Full retraining is often too costly to serve merely as an audit oracle. We formulate and solve a narrower, one-sided problem: when can deletion permutations themselves falsify a universal retrain-equivalence tolerance?

We treat two orders as the boundary paths of a noncommutative comparative statics (NCS) relation cell. For observed route outputs Y in any metric space, their Chebyshev radius is the greatest population lower bound on worst-route error obtainable from Y when the common target is otherwise unconstrained; half the diameter is an always-valid weaker bound. Consequently, an endpoint gap above 2ε rejects the claim that both routes are within ε of one target, without computing that target. Agreement cannot certify deletion: all routes can coincide arbitrarily far from retraining.

For random outputs we give a simultaneous, finite-sample MMD margin rule with family-wise error control. A Boolean-cube formulation identifies contextual exchange cells, and a trusted affine promise reduces each cell to finitely many basis probes. For a fixed-preconditioner relinearized ridge-deletion protocol, we derive the exact cell defect, its global commutation criterion, quadratic response order, and target-free retained-objective lower bound.

A reproducible 442-sample ridge case study checks all 97,461 deletion pairs, executes all six orders of a three-request set, and supplies a powered synthetic stochastic illustration. The artifact verifies the derived cell and case-study formulas and uses exact retraining only afterward as a validation oracle. The contribution is an unlearning-specific synthesis and audit method, not a new Chebyshev, kernel, commutator, or metamorphic-testing theorem.

Keywords. machine unlearning; oracle-free auditing; path dependence; metamorphic testing; Chebyshev radius; maximum mean discrepancy; noncommutative comparative statics.

Claim boundary. PC-Audit is a sound, incomplete *rejection* certificate for claims with one common reset target or law. It does not certify that data were forgotten. “Sharp” refers only to a fixed observed population output set, with no information about the target beyond membership in the declared metric space. Finite-sample bounds are conservative, and unobserved orders remain uncontrolled.

1 Introduction

Machine unlearning asks a trained system to remove the influence of selected records more cheaply than full retraining [Cao and Yang, 2015, Ginart et al., 2019, Bourtole et al., 2021]. Strong formulations compare the updated model or its law with the result of training on the retained data [Guo et al., 2020, Neel et al., 2021]. Approximate methods, however, are stateful numerical protocols. Residuals are recomputed, optimizer state is carried forward, preconditioners become stale, and stopping rules can depend on the path. Deleting i and then j may therefore differ from deleting j and then i , even though the retained dataset is identical.

Order dependence is not itself new. Sequential deletion guarantees and adaptive request streams are established topics [Neel et al., 2021, Gupta et al., 2021]; recent work directly studies forget-sample order in gradient-ascent unlearning [Kumar et al., 2026]; and training-stage history can obstruct path-oblivious retrain equivalence [Yu et al., 2025]. The question here is more operational:

Can route disagreement alone prove that a universal all-order retrain-equivalence tolerance is false, without constructing a retrained reference?

The elementary triangle inequality suggests a pairwise answer, but turning it into a useful audit requires several non-elementary choices: the quantifiers of the claim, the sharp multi-route bound, stochastic-law semantics, a level-controlled margin test, route failures, contextual rather than stateless updates, finite auditability, numerical lower-bound safety, and an explicit no-go boundary. We assemble these pieces as *permutation-cell auditing* (PC-Audit), an application of NCS to machine unlearning.

Contributions.

1. We identify the Chebyshev radius of a fixed observed route family as the sharp target-free population lower bound on its worst error to any otherwise unconstrained common reset target.
2. We give a simultaneous bounded-kernel MMD lower confidence rule that rejects a common-target tolerance with controlled family-wise error.
3. We present fixed deletion requests as a Boolean cube of contextual exchange cells and prove a complete finite test on a reachable affine hull under a trusted affine-map promise.
4. We solve one concrete protocol exactly: fixed-preconditioner, relinearized ridge deletion. Its cell defect has a closed form, exact quadratic amplitude scaling, and a target-free retained-objective certificate.
5. We provide a fully reproducible deterministic and stochastic case study, plus a no-go theorem showing why route agreement cannot positively certify unlearning.

Recognizable review communities. The paper sits at the intersection of machine unlearning and auditing, nonparametric kernel testing, numerical optimization, and metamorphic software testing. Each mathematical ingredient is standard enough to admit conventional review; the candidate novelty is the scoped synthesis and application.

2 Related work and positioning

Unlearning guarantees and approximations. Certified removal, deletion algorithms for convex models, SISA training, and approximate linear-model deletion establish retraining-based or distribution-based reference semantics [Guo et al., 2020, Neel et al., 2021, Bourtole et al., 2021, Izzo et al., 2021]. Influence functions motivate many local approximations [Koh and Liang, 2017]. PC-Audit does not propose a faster unlearning update. It treats an existing update implementation as a response protocol and tests a necessary consequence of a common-target claim.

Sequential and path-dependent unlearning. Gupta et al. [2021] study adaptive deletion sequences. Kumar et al. [2026] directly show that different orderings of the same forget samples can produce different outcomes in gradient-ascent unlearning. Yu et al. [2025] study dependence on earlier training-stage order. These works preclude any claim that deletion-order sensitivity or local commutators are new. Our fixed checkpoint and fixed deletion set instead support a target-independent *lower bound* against every common reset target.

Auditing and oracle limits. Thudi et al. [2022] show why output or parameter proximity alone cannot prove that data were absent from a training trajectory. Current work develops retraining-free verification models [Ye et al., 2026], representation-level diagnostics [Cosma and Finke, 2026], and validated selective screens while emphasizing limits of oracle-free certification [Yang and Yeung, 2026]. PC-Audit is deliberately weaker: rejection is logically sound for a precisely declared target tolerance, while non-rejection is inconclusive.

Metamorphic and kernel testing. Metamorphic testing evaluates relations among executions when a direct oracle is unavailable [Segura et al., 2016]. PC-Audit is an unlearning-specific metamorphic relation equipped with a metric target-error interpretation. MMD and kernel two-sample testing are established [Gretton et al., 2012]; current work includes equivalence margins [Liu and Gandy, 2026] and machine-unlearning-oriented divergence tests [Ribero et al., 2026]. We do not introduce a new kernel test. We use a simple simultaneous MMD lower confidence bound as input to the NCS geometric conversion.

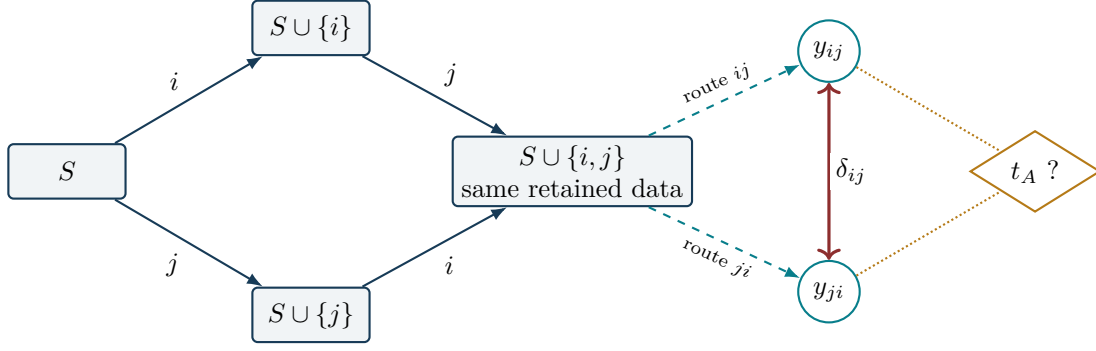


Figure 1: Permutation cell. The external square commutes because both paths delete the same records. The response endpoints can disagree. Their separation δ_{ij} lower-bounds the worst distance to every unknown common target by $\delta_{ij}/2$.

3 Audit semantics

Fix a finite, non-adaptively selected deletion set A , a family Π_A of permitted orders, and a starting checkpoint. Every $\pi \in \Pi_A$ has the same retained dataset $D \setminus A$, so the routes are externally equivalent. Their internal carried states may differ.

Let (X, d) be the declared comparison space. Its points can be model parameters, prediction vectors on a fixed probe set, functions modulo a gauge, or probability laws. A successful route π returns $y_\pi \in X$. A chosen reset semantics supplies one common target $t_A \in X$: for example, the output of a deterministic canonical trainer on $D \setminus A$, or the law of a randomized canonical trainer.

Definition 3.1 (Uniform all-order target claim). The response protocol satisfies a uniform ε -target claim on Π_A if every route is defined and

$$d(y_\pi, t_A) \leq \varepsilon \quad \text{for every } \pi \in \Pi_A.$$

Warm starts, cached state, stopping rules, and seed coupling belong to the response protocol. They are not silently equated with reset retraining. Route-conditioned, seed-coupled, or set-valued targets require a different claim and are outside [definition 3.1](#).

4 Sharp target-free geometry

For a nonempty finite set $Y = \{y_1, \dots, y_m\} \subset X$, define

$$r_X(Y) = \inf_{z \in X} \max_a d(y_a, z), \quad \text{diam}(Y) = \max_{a,b} d(y_a, y_b).$$

The infimum permits metric spaces in which no center is attained.

Theorem 4.1 (Sharp output-only certificate). *For every common target $t \in X$,*

$$\max_a d(y_a, t) \geq r_X(Y) \geq \frac{1}{2} \text{diam}(Y). \quad (4.1)$$

Moreover, $r_X(Y)$ is the greatest lower bound guaranteed from the fixed observed set Y alone when the target is otherwise unconstrained in X . Precisely, if $B(Y)$ satisfies

$$\max_a d(y_a, t) \geq B(Y) \quad \text{for every } t \in X,$$

then $B(Y) \leq r_X(Y)$.

Proof. Substituting $z = t$ into the infimum gives the first inequality. For every pair a, b and every z ,

$$d(y_a, y_b) \leq d(y_a, z) + d(y_b, z) \leq 2 \max_c d(y_c, z).$$

Maximize over pairs and infimize over z .

For optimality, suppose $B(Y) > r_X(Y)$. By the definition of an infimum, some $z \in X$ satisfies $\max_a d(y_a, z) < B(Y)$, contradicting the proposed guarantee at target $t = z$. $\square$

Corollary 4.2 (Pair-cell rejection). *If $d(y_\pi, y_\sigma) > 2\varepsilon$, no common target lies within ε of both routes. Hence the uniform ε -target claim is false.*

If independent information restricts t to a nonempty $T \subseteq X$, the stronger quantity is $\inf_{z \in T} \max_a d(y_a, z)$. The sharpness claim in [theorem 4.1](#) assumes no such information. Likewise, an observed-route radius does not control unobserved orders.

4.1 Hilbert decomposition

Let X be a real Hilbert space, with two outputs y_+ and y_- . Define

$$c = y_+ - y_-, \quad m = \frac{y_+ + y_-}{2}.$$

Proposition 4.3 (Two-route audit identity). *The unique two-point minimum-enclosing-ball center is m , with radius $\|c\|/2$. For every target t ,*

$$\frac{\|y_+ - t\|^2 + \|y_- - t\|^2}{2} = \|m - t\|^2 + \frac{\|c\|^2}{4}. \quad (4.2)$$

Proof. Apply the parallelogram identity to $y_\pm - t = (m - t) \pm c/2$. The average of the two squared distances is at least $\|c\|^2/4$, while m attains equal distances $\|c\|/2$. $\square$

The midpoint removes the antisymmetric route component, but $\|m - t\|$ can remain arbitrarily large. This distinction is fundamental.

Proposition 4.4 (Zero-defect no-go). *No finite-valued function of route disagreement alone can uniformly upper-bound reset-target error. For every $M > 0$, there is a protocol whose routes all agree while their common output is distance M from the reset target.*

Proof. Take $X = \mathbb{R}$, target $t = 0$, and every deletion map to be the constant map $x \mapsto M$. Every route has the same output and all relation-cell defects vanish, while target error equals M . $\square$

Thus PC-Audit can falsify a target claim when its lower bound crosses the declared tolerance. It cannot certify the claim when the bound is small.

5 Stochastic route laws

Let route π return a probability law P_π on $\mathcal{Z}$, and let the selected reset law be Q_A . Randomness semantics must match:

- Cloning one trained checkpoint estimates conditional laws $P_{\pi|W=w_0}$. It can audit only a matching conditional reset semantics.
- An end-to-end claim about the algorithmic marginal law requires independent reruns of both original training and unlearning.

A common-seed coupled difference is not by itself a difference between marginal laws. Random route failure must either be represented by a distinguished output symbol or audited with a separate failure-probability tolerance. One crash rejects only an almost-sure availability claim.

5.1 A simultaneous MMD margin certificate

Let k be a characteristic positive-definite kernel with feature map $\phi : \mathcal{Z} \rightarrow \mathcal{H}_k$ and global diagonal bound

$$k(z, z) = \|\phi(z)\|_{\mathcal{H}_k}^2 \leq K.$$

For law P , let $\mu_P = \mathbb{E}_P \phi(Z)$, so that $\text{MMD}_k(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}$ [Gretton et al., 2012].

Suppose s routes are audited and route a supplies n_a independent complete-route replicates $Z_{a1}, \dots, Z_{an_a}$. Cross-route dependence, including paired common random numbers, is allowed; within-route replicates must be independent. Define

$$\hat{\mu}_a = \frac{1}{n_a} \sum_{\ell=1}^{n_a} \phi(Z_{a\ell}),$$

and, for $0 < \delta < 1$,

$$e_a(\delta) = \sqrt{\frac{K}{n_a}} + \sqrt{\frac{2K \log(s/\delta)}{n_a}}. \quad (5.1)$$

Theorem 5.1 (Simultaneous MMD rejection margin). *With probability at least $1 - \delta$, simultaneously for all routes,*

$$\|\hat{\mu}_a - \mu_a\|_{\mathcal{H}_k} \leq e_a. \quad (5.2)$$

Consequently, every pair obeys

$$\text{MMD}_k(P_a, P_b) \geq L_{ab} := [\|\hat{\mu}_a - \hat{\mu}_b\| - e_a - e_b]_+. \quad (5.3)$$

If $\hat{r}$ is the minimum-enclosing-ball radius of the empirical mean embeddings, then the population radius satisfies

$$r_{\mathcal{H}_k}(\{\mu_a\}) \geq L_{\text{rad}} := \left[\hat{r} - \max_a e_a \right]_+. \quad (5.4)$$

Therefore either condition

$$\max_{a,b} L_{ab} > 2\varepsilon, \quad \text{or} \quad L_{\text{rad}} > \varepsilon \quad (5.5)$$

rejects, with family-wise error at most δ , the claim that every route law lies within MMD distance ε of one common reset law.

Proof. For one route, Jensen's inequality and independence give

$$\mathbb{E} \|\hat{\mu}_a - \mu_a\| \leq \sqrt{\mathbb{E} \|\hat{\mu}_a - \mu_a\|^2} \leq \sqrt{K/n_a}.$$

Changing one replicate changes the empirical mean, and hence the norm of its error, by at most $2\sqrt{K}/n_a$. McDiarmid's bounded-difference inequality [McDiarmid, 1989] yields

$$\mathbb{P} \left[\|\hat{\mu}_a - \mu_a\| > \sqrt{K/n_a} + t \right] \leq \exp \left(-\frac{n_a t^2}{2K} \right).$$

Set $t = \sqrt{2K \log(s/\delta)/n_a}$ and union-bound over routes to prove (5.2). The reverse triangle inequality gives (5.3).

On the event (5.2), corresponding empirical and population embeddings are within $\eta = \max_a e_a$. A finite-set minimum-enclosing-ball radius is 1-Lipschitz under such a Hausdorff perturbation: the same center enlarges a ball by at most η in either direction. This proves (5.4). Theorem 4.1 completes the conversion to (5.5). $\square$

Implementation safeguards. The statistic in theorem 5.1 is the RKHS norm of empirical means, often called biased MMD, not the possibly negative unbiased estimator of MMD^2 . Kernel, bandwidth, routes, tolerance, and transformations must be fixed independently or handled by a valid selection procedure. A predeclared finite kernel family can receive a multiplicity correction; continuous data-adaptive optimization needs sample splitting or a uniform bound. Approximate minimum-enclosing-ball software must return a certified lower bound or subtract a proven optimization error before rejection.

6 Deletion orders as a Boolean cube

Let $A = \{1, \dots, q\}$. External vertices are subsets $S \subseteq A$. For $i \notin S$, an edge $S \rightarrow S \cup \{i\}$ processes request i . For $i, j \notin S$, the contextual exchange square is

$$\left(S \xrightarrow{i} S \cup \{i\} \xrightarrow{j} S \cup \{i, j\} \right) \Rightarrow \left(S \xrightarrow{j} S \cup \{j\} \xrightarrow{i} S \cup \{i, j\} \right). \quad (6.1)$$

There are $\binom{q}{2} 2^{q-2}$ such squares for $q \geq 2$.

Let response edges be context-dependent partial maps

$$U_i^S : X_S \rightarrow X_{S \cup \{i\}}.$$

Equal outputs are not enough when one route is undefined; equality of partial maps requires both equal domains and equal values.

Proposition 6.1 (Contextual-square completeness). *If the two composites on every contextual square have equal domains and equal values, then every pair of monotone deletion paths with common endpoints induces the same partial response map.*

Proof. Any two permutations differ by adjacent transpositions. Each adjacent transposition is one contextual square. Equality of partial maps is preserved by composition, so a transposition chain transforms either path into the other without changing the response map. $\square$

For stateless total maps on one space, pairwise commutation is necessary and sufficient for equality at every subset endpoint, or equivalently for all request words up to permutation. It need not be necessary when only the one full-set endpoint is checked: a noncancellative suffix can mask an earlier defect. Approximate cells also require Lipschitz filling bounds, because a suffix can amplify each local residual.

6.1 Finite affine cell tests

Suppose one contextual cell has a common affine domain and affine defect

$$\Omega(\theta) = U_j^{S \cup \{i\}} U_i^S(\theta) - U_i^{S \cup \{j\}} U_j^S(\theta).$$

Let $\theta_0, \dots, \theta_r$ be an affine basis of an r -dimensional reachable affine hull L . Define linear maps $V, W : \mathbb{R}^r \rightarrow \mathbb{R}^d$, with V injective, by

$$Va = \sum_{k=1}^r a_k (\theta_k - \theta_0), \quad Wa = \sum_{k=1}^r a_k (\Omega(\theta_k) - \Omega(\theta_0)).$$

Then

$$\Omega(\theta_0 + Va) = \Omega(\theta_0) + Wa. \quad (6.2)$$

Proposition 6.2 (Affine-basis completeness). *The cell commutes on L if and only if its defect vanishes on one affine basis of L . On $\Delta = \text{conv}\{\theta_0, \dots, \theta_r\}$,*

$$\sup_{\theta \in \Delta} \|\Omega(\theta)\| = \max_{0 \leq a \leq r} \|\Omega(\theta_a)\|. \quad (6.3)$$

Proof. An affine map is determined by its affine-basis values, proving (6.2) and the first claim. For $\theta = \sum_a \lambda_a \theta_a \in \Delta$, convexity of the norm gives

$$\|\Omega(\theta)\| \leq \sum_a \lambda_a \|\Omega(\theta_a)\| \leq \max_a \|\Omega(\theta_a)\|.$$

The reverse inequality holds because the vertices belong to Δ . $\square$

This finite test requires a trusted affine-map promise, a common valid domain containing the hull, exact evaluations, and state-injection access. Basis agreement cannot establish affinity. Noisy probes need a separate conditioning-aware statistical analysis.

7 Exact ridge relation-cell solution

Consider the fixed-normalization ridge objective

$$F_D(\theta) = \frac{1}{2} \sum_{k \in D} (x_k^\top \theta - y_k)^2 + \frac{\lambda}{2} \|\theta\|^2, \quad \lambda > 0. \quad (7.1)$$

Write

$$H = \lambda I + \sum_{k \in D} x_k x_k^\top, \quad b = \sum_{k \in D} y_k x_k, \quad \theta_D = H^{-1} b.$$

Fix a symmetric positive-definite preconditioner P . The fixed-preconditioner relinearized request update is

$$U_i^\tau(\theta) = \theta + \tau P x_i (x_i^\top \theta - y_i), \quad 0 \leq \tau \leq 1. \quad (7.2)$$

It reevaluates request i 's residual at the carried state while retaining the same preconditioner.

Let $r_i(\theta) = x_i^\top \theta - y_i$ and $\alpha_{ij} = x_j^\top P x_i$. For a fixed start state, call p the response order when a nonzero defect norm has leading scaling $C\tau^p + o(\tau^p)$ as $\tau \downarrow 0$, with $C > 0$.

Theorem 7.1 (Exact ridge cell defect). *For distinct requests i, j ,*

$$\boxed{U_j^\tau U_i^\tau(\theta) - U_i^\tau U_j^\tau(\theta) = \tau^2 \alpha_{ij} [P x_j r_i(\theta) - P x_i r_j(\theta)]}. \quad (7.3)$$

For $\tau \in (0, 1]$, nonzero features, and positive-definite P , the two affine maps commute at every state if and only if either

$$x_i^\top P x_j = 0, \quad (7.4)$$

or there is a nonzero scalar c with

$$(x_j, y_j) = c(x_i, y_i). \quad (7.5)$$

At $\tau = 0$, both maps are identities. At a start state independent of τ , a nonzero coefficient in (7.3) gives exact NCS response order two: the defect scales exactly as τ^2 .

Proof. Because

$$r_j(U_i^\tau \theta) = r_j(\theta) + \tau x_j^\top P x_i r_i(\theta),$$

direct substitution in the two composites leaves (7.3). If $\alpha_{ij} = 0$, the defect vanishes. Otherwise the affine bracket vanishes for every θ exactly when

$$x_j x_i^\top - x_i x_j^\top = 0, \quad x_j y_i - x_i y_j = 0,$$

after multiplying by P^{-1} . Nonzero real vectors satisfy the first identity precisely when $x_j = c x_i$; the second then gives $y_j = c y_i$. The converse and the quadratic scaling are immediate. $\square$

7.1 Target-free retained-objective certificate

Write $U_i = U_i^1$, define

$$z_{ij} = U_j U_i(\theta_D), \quad z_{ji} = U_i U_j(\theta_D),$$

and let θ_{-ij} be the exact ridge minimizer after deleting i, j . The retained Hessian is

$$H_{-ij} = H - x_i x_i^\top - x_j x_j^\top \succeq \lambda I.$$

Corollary 7.2. *In every norm,*

$$\max\{\|z_{ij} - \theta_{-ij}\|, \|z_{ji} - \theta_{-ij}\|\} \geq \frac{1}{2} \|z_{ij} - z_{ji}\|. \quad (7.6)$$

Moreover,

$$\begin{aligned} \max_{\pi} \{F_{-ij}(z_{\pi}) - F_{-ij}(\theta_{-ij})\} &\geq \frac{1}{8} \|z_{ij} - z_{ji}\|_{H_{-ij}}^2 \\ &\geq \frac{\lambda}{8} \|z_{ij} - z_{ji}\|_2^2. \end{aligned} \quad (7.7)$$

Proof. Equation (7.6) is [theorem 4.1](#) for two points. Since

$$F_{-ij}(z) - F_{-ij}(\theta_{-ij}) = \frac{1}{2} \|z - \theta_{-ij}\|_{H_{-ij}}^2,$$

apply (7.6) in the H_{-ij} -norm and use $H_{-ij} \succeq \lambda I$. $\square$

For $m = (z_{ij} + z_{ji})/2$, the exact objective decomposition is

$$\frac{1}{2} \sum_{\pi \in \{ij, ji\}} [F_{-ij}(z_\pi) - F_{-ij}(\theta_{-ij})] = \frac{1}{2} \|m - \theta_{-ij}\|_{H_{-ij}}^2 + \frac{1}{8} \|z_{ij} - z_{ji}\|_{H_{-ij}}^2. \quad (7.8)$$

The last term is the identifiable antisymmetric order contribution. The first is target-dependent and can remain large after route symmetrization.

Exact reset comparator. For batch feature matrix X_A and label vector y_A , standard Sherman–Morrison–Woodbury algebra [[Golub and Van Loan, 2013](#)] gives

$$\theta_{-A} = \theta_D + H^{-1} X_A (I - X_A^\top H^{-1} X_A)^{-1} (X_A^\top \theta_D - y_A), \quad (7.9)$$

when the retained Hessian is positive definite. This exact set-dependent reset commutes. It is used below only as a post-audit validation oracle. Likewise, freezing all residual vectors produces commuting translations that can still be wrong, illustrating [proposition 4.4](#).

8 Solved case study

8.1 Design and reproducibility

We use the 442-sample, ten-feature diabetes dataset bundled with scikit-learn [[Pedregosa et al., 2011](#)]. The target is population standardized, and $\lambda = 0.05$. The study is mathematical and non-clinical: it makes no predictive or population-generalization claim.

The protocol uses $P = H^{-1}$ in (7.2). We exhaustively screen all $\binom{442}{2} = 97,461$ record pairs by the analytic cell formula and direct composition. The screen selects a strong transparent witness before stochastic samples are generated. It is exploratory, not an estimate of deployed prevalence.

The artifact pins NumPy 2.5.1, SciPy 1.18.0, and scikit-learn 1.9.0; records the dataset-array, script, and requirements SHA-256 digests; fixes seed 20260728; and resolves its output path from the script location. All formulas, checks, and values below are generated by one inspectable script.

8.2 Two-route deterministic witness

The strongest Euclidean cell uses records 32 and 322. Across all 97,461 pairs, the maximum discrepancy between analytic and directly composed defects is 2.26×10^{-15} . For the selected pair,

$$\|z_{32,322} - z_{322,32}\|_2 = 0.0272943461,$$

and in the retained-Hessian norm,

$$\|z_{32,322} - z_{322,32}\|_{H_{-32,-322}} = 0.0115529959.$$

Without using the reset target, [corollary 7.2](#) therefore gives

$$\max_{\pi} \|z_{\pi} - \theta_{-32,-322}\|_{H_{-32,-322}} \geq 0.00577649793 \quad (8.1)$$

and

$$\max_{\pi} [F_{-32,-322}(z_{\pi}) - F_{-32,-322}(\theta_{-32,-322})] \geq 1.66839642 \times 10^{-5}. \quad (8.2)$$

Table 1: Target-free quantities versus post-audit validation. Target-free columns use only route outputs and declared metric information. Deterministic rows use an exact target computed afterward; the stochastic row uses the closed-form population route radius and defines no reset law.

Audit	Quantity	Target-free bound	Post-audit validation
Pair {32, 322}	Worst parameter error, retained- H norm	0.00577650	0.01073861
Pair {32, 322}	Worst retained objective excess	1.6684×10^{-5}	5.7659×10^{-5}
Triple {32, 322, 141}	Worst parameter error, retained- H norm	0.01110879	0.03006635
Stochastic pair	Route-law MMD radius	0.23853627	population radius 0.30764560

Only after computing the target-free quantities does the script solve the exact retained-data solution. The two actual parameter errors are 0.0102969 and 0.0107386 in the retained-Hessian norm, while the objective excesses are 5.30130×10^{-5} and 5.76588×10^{-5} . The Pythagorean decomposition is

$$5.53359 \times 10^{-5} = 3.86520 \times 10^{-5} + 1.66840 \times 10^{-5},$$

to 6.61×10^{-19} absolute numerical error. Thus 30.15% of mean two-route objective excess is the exactly identifiable antisymmetric order component.

8.3 Six routes and affine reconstruction

For deletion set {32, 322, 141}, all six orders are executed from the same checkpoint. Their retained-Hessian diameter is 0.0222175847, so half the diameter gives the target-free lower bound 0.0111087924. Exhaustive floating-point support enumeration returns a minimum-enclosing-ball radius equal to this value within 10^{-15} ; the actual worst target error, computed only for validation, is 0.0300663494. The bound recovers 36.95% of that witness’s worst error. The diameter-supporting orders are (322, 32, 141) and (141, 32, 322).

Because the ridge protocol is affine, one base point and ten coordinate probes reconstruct the full cell defect. The linear-part Frobenius error is 4.26×10^{-14} , and the constant-part Euclidean error is 1.39×10^{-13} . One thousand random simplex probes respect the vertex bound (6.3). Sampling illustrates rather than proves that bound; the proof is [proposition 6.2](#).

8.4 Response order and stochastic illustration

At amplitudes

$$\tau \in \{1, 1/2, 1/4, 1/8, 1/16, 1/32\},$$

the Euclidean defects are

τ	1	1/2	1/4	1/8	1/16	1/32
$\ c_\tau\ _2$	2.7294×10^{-2}	6.8236×10^{-3}	1.7059×10^{-3}	4.2647×10^{-4}	1.0662×10^{-4}	2.6655×10^{-5}

and the fitted log–log slope is 1.999999999997464.

For a deliberately powered stochastic illustration, we add fresh independent isotropic Gaussian noise to each complete route in retained-Hessian coordinates. Noise scale and Gaussian-RBF bandwidth are effect-calibrated from the deterministic witness, then fixed before random-number generation. This is not a prospective power study.

With 3,000 replicates per route, $K = 1$, $\delta = 0.05$, and declared $\varepsilon = 0.10$,

$$\widehat{\text{MMD}} = 0.612769, \quad e_a = 0.0678483, \quad L_{12} = 0.477073.$$

Because $L_{12} > 2\varepsilon = 0.20$, [theorem 5.1](#) rejects the common conditional reset-law tolerance with family-wise error at most 0.05. Equivalently for two routes, the radius lower bound 0.238536 exceeds ε . The Gaussian closed-form MMD is 0.615291; its agreement with the realized lower bound is a one-seed consistency check, not an empirical coverage estimate. Coverage follows from [theorem 5.1](#).

9 Operational PC-Audit protocol

PC-Audit.

1. Declare one common reset target or law, comparison metric, tolerance, route family, and randomness semantics.
2. Fix a non-adaptive deletion set. Clone one checkpoint only for a deterministic or conditional-on-checkpoint audit; rerun original training for an end-to-end marginal-law audit.
3. Execute externally equivalent deletion orders. A deterministic route failure rejects the all-order availability claim. In a stochastic audit, include failure in the output law or audit its probability separately.
4. Deterministic case: compute a certified Chebyshev-radius lower bound, or use half the observed diameter. Reject only if it exceeds ε .
5. Stochastic case: compute (5.3) and, when numerically safe, (5.4). Reject only under (5.5).
6. Under a trusted affine promise and state-injection access, reconstruct tested cell defects from affine-basis probes. Full path consistency requires every relevant contextual square and its domain.
7. If no rejection occurs, report **inconclusive**, never “certified unlearned.”

Complexity. Two selected orders require two route executions. A full stateless pair audit uses $\binom{q}{2}$ cells, whereas a context-dependent Boolean cube contains $\binom{q}{2}2^{q-2}$ cells. The full family can therefore be exponentially large. Audit design must choose whether to test a specified small route set, a risk-ranked sample, or a structural promise such as statelessness.

Metric choice. Parameter distance can be distorted by representation symmetries. Prediction vectors on a fixed probe set or output laws may better match operational claims, but probe selection and kernel bandwidth then enter the audit specification. MMD detects differences in the declared output law; it is not a privacy or deletion semantic by itself.

10 Limitations and falsification conditions

1. **One-sidedness is unavoidable from route geometry alone.** [Proposition 4.4](#) rules out positive certification.
2. **A common target is essential.** Route-conditioned, seed-coupled, or set-valued reset semantics require different geometry.
3. **Observed routes are the scope.** A sharp radius on sampled routes says nothing about unobserved permutations without structural assumptions.
4. **Context can be exponential.** Pairwise stateless commutation does not replace contextual cube cells in a genuinely state-dependent system.
5. **Numerical lower bounds must be safe.** An approximate enclosing-radius overestimate can be anti-conservative. The case study uses half the diameter as the operative bound and treats its floating-point MEB as a diagnostic.
6. **The case study is not a production benchmark.** It proves that the method can extract a nontrivial target-free certificate in an inspectable problem. Deep-model operational validation remains future work.

The candidate novelty would be falsified by prior work that already combines same-deletion-set permutations, the sharp common-target radius conversion, a level-controlled stochastic margin rule, and the explicit one-sided unlearning semantics. The mapped literature contains each ingredient or nearby audit objective separately, but we found no such combined construction. This is a positioning result, not proof of priority.

11 Discussion

The practical gain is not that a triangle inequality is difficult. It is that the NCS presentation exposes a cheap, target-free observable with a precise meaning. A route gap is no longer only “instability.” It is a lower bound on the failure of a universal common-target claim. The Boolean cube says which executions are comparable; partial-map semantics preserve directional failures; the Chebyshev radius solves the multi-route geometry; the stochastic bound matches probabilistic claims; and the no-go theorem prevents consistency from being marketed as forgetting.

The ridge witness shows an additional benefit of the field-level view. The cell defect separates into overlap, $\alpha_{ij} = x_j^\top P x_i$, and residual-weighted directions. It predicts exact quadratic response order, identifies global commutation conditions, and converts directly to retained-objective debt. These are actionable diagnostics: an operator can rank request pairs by predicted defect, distinguish orthogonality from duplicate labeled records, and quantify an unavoidable worst-route error before paying for reset retraining.

The broader research direction is to develop relation-cell libraries for actual unlearning protocols, safe adaptive route selection, interval-verified radius solvers, function-space metrics invariant to parameter gauge, and context-covering designs with explicit power and cost.

12 Conclusion

Permutation-cell auditing solves a concrete oracle-free problem for sequential machine unlearning. If externally equivalent deletion orders separate by more than twice a declared tolerance, then for every proposed common reset target, at least one route lies outside tolerance. The observed Chebyshev radius is the sharp unrestricted population form of that statement; a bounded-kernel MMD margin provides a finite-sample stochastic version. Boolean-cube and affine structure specify when finite cell tests control larger path families, and an exact ridge example demonstrates nontrivial target-free parameter and objective certificates.

The conclusion is deliberately asymmetric: disagreement can prove failure; agreement cannot prove forgetting. That limitation is not a weakness hidden by the method. It is the central fact that makes the method auditable.

Artifact availability and research conduct

The accompanying artifact contains the checkpoint proofs, adversarial reviews, literature collision map, pinned environment, executable case study, JSON results, source manuscript, and rendered PDF. No external individual was contacted during the research or review process; the recorded reviews were performed by independent internal adversarial agents.

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